

Student Name: Period: Date: ★ 1-Page Individual Defense

PHASE 1 SELECT YOUR INERTIA REGIME & CODENAME

1. Inertia of Rest: Stationary mass stubbornly stays at rest when support is pulled fast (e.g. Tablecloth slip, coin-card flick, embroidery hoop drop).
2. Inertia of Motion: Moving mass stubbornly continues at constant velocity when carrier halts (e.g. Cart barrier crash, curb stop, rolling launch).
3. Inertia of Direction: Rotating mass flies off along straight tangent line when force is cut (e.g. Whirling string cut, pie-pan exit gap, wheel throw).

Demonstration Title / Codename:

PHASE 2 APPARATUS & MATERIALS INVENTORY

1. Inertial Mass (m): 2. Support / Carrier Surface:
3. Applied Force Trigger: 4. Safety Catch Zone / Basin:

PHASE 3 ANNOTATED SETUP & ACTION BLUEPRINT DIAGRAM

✓ Draw apparatus BEFORE action ✓ Label Inertial Mass (m) & Surface ✓ Arrow for Applied Force Trigger ✓ Dashed arrow for Mass Path
Draw neat, labeled vectors with a ruler · Grid: 12 mm

PHASE 4 EXECUTION PROTOCOL (STEP-BY-STEP PROCEDURE)

1. Setup & Alignment: How is apparatus positioned before action?
2. The Trigger (Quick Action): What exact fast action do you execute?
3. The Result (Mass vs Support): What does support do vs. inertial mass?

PHASE 5 FREE-BODY DIAGRAM (FBD) DEFENSE: NET FORCE ANALYSIS

Diagram A: Before Action (Static / Constant v) F_net = 0 N (Equilibrium)
Diagram B: During Split-Second Action Horizontal Force ≈ 0 N
Draw balanced vertical forces (FN, Fg)
Draw horizontal forces acting ON THE MASS
Misconception Buster: Inertia is NOT a force! There is ZERO forward force pushing the mass.

PHASE 6 SCIENTIFIC DEFENSE (CER) & REAL-WORLD CONNECTION

Scientific Explanation (Claim & Reasoning): Explain using: Mass, Inertia, Friction, and Net External Force (F_net = 0).
Real-World System (e.g. Seatbelt, Airbag, Cargo): How does Newton's 1st Law protect humans or explain everyday motion?

DEFENSE RUBRIC: [] Apparatus & Safety (4) [] Blueprint Diagram (4) [] FBD Accuracy (4) [] CER Defense (4) SCORE: / 16 pts Teacher: